

When Models Answer Without Seeing: Human-Calibrated Diagnosis of Linguistic Prior Exploitation in Vision-Language Models

Anonymous submission

Abstract

Analogous to hypothesis-only baselines that exposed annotation artifacts in Natural Language Inference (Poliak et al. 2018), we investigate how much of VQA behavior is explainable without any visual information. Across multiple open-source Vision-Language Models (VLMs), blind inference reveals characteristic answer distribution biases shaped by the training distribution (McCoy, Frank, and Linzen 2024): models collapse to “no” for yes/no questions and “0” for count questions at rates far exceeding human baselines. To characterize how these priors depend on lexical cues, we introduce *control question variants*: four progressively weaker reformulations that remove entity anchors while preserving question type. A single instruction further reveals a spectrum from *instruction-gated* prior use to unconditional hallucination. Blind outputs are also highly overconfident: 80–90% of tokens have high mean log-probability, a calibration failure invisible to accuracy-only evaluations. Using a human blind-answer dataset (N=36, 113 questions, IRB-approved), we show that the model group closest to human blind priors is not the full VLM but the VLM backbone decoder: removing the vision encoder yields the highest human-model semantic agreement, while full VLMs and standalone LLMs exhibit distinct failure regimes. Question-type analysis then reveals where this overlap comes from: yes/no and action questions show strong human-model prior sharing, while count and grounded-reference questions are strongly anti-aligned.

1 Introduction

Vision-Language Models (VLMs) achieve impressive performance on visual question answering (VQA) benchmarks (Antol et al. 2015; Goyal et al. 2016), yet a basic question remains unanswered: how much of this performance reflects genuine visual grounding, and how much can be explained by linguistic priors inherited from large-scale language model pretraining?

This question parallels a well-known episode in Natural Language Inference (NLI): Poliak et al. (2018) showed that a model trained on hypotheses alone—ignoring the corresponding premises—substantially exceeded the random baseline across ten NLI datasets. The finding revealed annotation artifacts embedded in the benchmarks themselves. We apply the same diagnostic logic to VQA: the *image* plays the role of the premise, and the question is the hypothesis. A model that answers correctly from the question alone is

exploiting a hypothesis-only shortcut.

McCoy, Frank, and Linzen (2024) provide the mechanistic explanation: large language models are shaped by their training objective over internet-scale text, and therefore produce outputs proportional to the probability of sequences in the training corpus. When a VQA question has a high-frequency answer in training data (“What color is the sky?” → “blue”), a VLM will produce that answer reliably regardless of the image. Blind VQA accuracy thus directly measures corpus frequency exploitation.

In this work, we conduct a diagnostic study of this phenomenon across multiple open-source VLMs. The paper is organized around three questions. First, how much blind accuracy can models obtain at scale, and what answer distribution biases underlie that performance? Second, how does prior-driven behavior change when we remove lexical anchors or explicitly instruct the model to answer from a plausible imagined scene? Third, how similar are blind model behaviors to human blind reasoning at the levels of correctness, answer distribution, and semantic agreement?

To answer these questions, we combine blind evaluation on 1,000 VQA questions, a control-variant perturbation suite, logprob-based confidence analysis, and a human blind-answer dataset (N=36 participants, 113 questions). The goal is not to claim that blind answering is inherently illegitimate; the goal is to identify *which* priors are shared with humans and *which* are model-specific artifacts—a distinction that prior accuracy-centric diagnostics cannot make.

Main findings.

1. Models answer a substantial fraction of VQA questions correctly without any image input, confirming that blind prior exploitation is pervasive rather than anecdotal.
2. VLM backbone decoders (language backbone without the vision encoder) are the model group most closely aligned with human blind priors. They reach the highest human-model semantic agreement, while full VLMs are lower and standalone LLMs diverge further.
3. Full VLMs and standalone LLMs do not simply share a generic “model prior.” They exhibit different failure regimes: full VLMs show stronger no-bias and zero-bias, while standalone LLMs are more heterogeneous and often less human-aligned in answer semantics.

4. Human-model prior alignment splits by question type: yes/no and action questions show strong overlap, while count and grounded-reference questions are anti-aligned. This identifies which priors are genuinely human-like and which are model-specific artifacts.
5. Models exhibit characteristic null-hypothesis biases under blind conditions: 62% of VLM yes/no answers are “no” (vs. human 33%), 70% of count answers are “0” (vs. human 5%), and color answers cluster on “black”. These biases are model-specific—not shared with human blind reasoners.
6. Blind outputs are severely overconfident: 80–90% of outputs have high mean token log-probability, and the instruction further raises confidence while reducing abstention, indicating more committed hallucination rather than more grounded reasoning.

Methodological contributions. Beyond the empirical findings, this work introduces two reusable diagnostic tools.

(i) **A semantic control-question extraction pipeline.** For each VQA question, we use GPT-4.1-mini with a structured JSON schema to automatically generate four progressively weaker reformulations: deictic markers removed, object noun replaced by a neutral placeholder, object replaced by its one-level-up hypernym, and content nouns fully pronominalized. This forms a controlled degradation axis that isolates *which lexical elements* carry prior information, independently of accuracy. The pipeline is fully specified (prompt and schema in Appendix C) and applicable to any VQA benchmark.

(ii) **An operation–entity taxonomy for VQA prior analysis.** We annotate each question with a semantic *operation type* (yes/no-existence, count, attribute, action, world knowledge, spatial, identity, temporal, text/OCR, comparison, causal) and an *entity type* (person, animal, object, food, vehicle, place, etc.). This taxonomy enables fine-grained diagnosis: yes/no and action questions show near-perfect human–model prior alignment (Jaccard ≈ 0.93), while count and OCR questions are anti-aligned ($r = -0.39$)—a distinction invisible at the aggregate level. Together, the two tools provide a principled diagnostic vocabulary for studying linguistic prior exploitation in any multimodal benchmark.

2 Related Work

Hypothesis-Only and Shortcut Baselines. Poliak et al. (2018) demonstrated that NLI classifiers achieve non-trivial accuracy using only the hypothesis, exposing annotation artifacts. Gururangan et al. (2018) further characterized these artifacts through systematic analysis of annotation bias. In VQA, Agrawal et al. (2018) showed that models relying on question-answer co-occurrence statistics could achieve high accuracy without visual understanding. Our work extends this diagnostic tradition by providing a human reference dataset alongside model evaluations, enabling calibrated measurement of the gap.

LMs as Corpus Frequency Models. McCoy, Frank, and Linzen (2024) showed that LLM failure patterns are predicted by three factors: the probability of the task, the in-

put, and the output in the training corpus. Models are biased toward high-probability outputs regardless of task requirements. We operationalize this insight: blind VQA accuracy directly measures the probability of question-answer pairs in the pretraining distribution.

VLM Hallucination and Shortcut Learning. Hallucination in VLMs—confident generation of visually ungrounded content—has been widely documented (Ji et al. 2023; Rohrbach et al. 2018; Xiao and Wang 2021). Chen et al. (2024) proposed MMStar to audit visual grounding by filtering out questions solvable without images. VLind-bench (Lee et al. 2024) specifically measures language prior influence across diverse benchmarks. Han et al. (2024) identified visual shortcuts through spurious image features. Complementarily, ? show that VLMs fail at fine-grained visual discrimination even when images are present—together with our work, bounding the visual grounding gap from both directions: models answer without images when they should not, and fail to use images when they must.

Human Cognition Under Uncertainty. Human reasoning under missing information is well characterized by dual-process theory (Tversky and Kahneman 1974; ?): fast, heuristic System 1 thinking relies on representativeness and availability, while System 2 engages deliberate reasoning when cues are insufficient. Studying how humans answer questions without visual evidence thus belongs to a well-established research tradition in psycholinguistics and cognitive modeling—examining what mechanisms govern speculation under uncertainty (?). Critically, humans do not share the corpus-frequency bias of pretrained LLMs: their yes-bias (67% “yes”) reflects pragmatic plausibility rather than training-distribution statistics, making human blind behavior the natural reference for what *linguistically natural* uncertainty looks like.

3 Experimental Setup

3.1 Models and Datasets

We evaluate open-source VLMs from the Qwen3-VL (Bai et al. 2025) and LLaVA (Liu et al. 2023, 2024) families, plus Qwen3 (Yang et al. 2025) LLMs as text-only baselines. The main blind-performance table reports the model subset for which the current zero-shot results are fully assembled; additional instruction-only comparisons are reported separately in Appendix E. All models are evaluated in zero-shot settings with temperature 0.

Our primary evaluation dataset consists of 1,000 questions sampled from the VQA v2.0 validation split (Goyal et al. 2016), spanning yes/no, numerical, and open-ended question types. For the human alignment analysis (Section 6), we use the 113-question subset on which the human study was conducted.

3.2 Blind Inference Conditions

We evaluate models under three conditions: (1) **Standard:** the original image with the question; (2) **Blind:** a blank 224×224 white PNG replacing the original image; and (3)

Blind+Inst: blank image with an additional instruction appended to the question: “*Note: No images are provided. For each question, imagine an appropriate image exists and answer based on the most common or universal scenario.*”

3.3 Control Question Variants and Semantic Taxonomy

As a methodological contribution, we introduce two annotation layers applied to all 1,000 VQA questions.

Control question extraction. To characterize *how* models exploit linguistic priors, we generate four progressively weaker reformulations of each question using GPT-4.1-mini (temperature 0, seed 42, JSON schema mode) for reproducibility:

1. **deictic_removed:** Deictic markers (*this, that, here, there*) and spatial qualifiers (left/right/top/bottom) are removed while preserving fluency. Tests whether spatial anchoring drives shortcut behavior.
2. **object_removed:** The specific object noun is replaced with the neutral placeholder “object.” Tests the contribution of entity identity to prior exploitation.
3. **weaker_object:** The specific noun is replaced with a one-level-up hypernym (dog → animal, car → vehicle, shirt → clothing). Tests whether category-level identity is sufficient to trigger prior recall.
4. **pronominalized:** All content noun phrases are replaced with minimal pronouns following grammar-aware rules (“What color is the dog?” → “What color is it?”; “How many cats are there?” → “How many are there?”). This maximally removes entity-specific prior cues while preserving question structure.

The degradation chain from original → pronominalized forms a monotonically decreasing axis of lexical prior cues, enabling a controlled measurement of how much each entity reference contributes to shortcut exploitation. The GPT extraction prompt and JSON schema are provided in Appendix C.

Operation–entity taxonomy. Each question is also annotated by the same GPT pipeline with two semantic dimensions: (1) *operation type*—the question’s cognitive demand: yes/no-existence (*exist*), counting (*count*), attribute identification (*attr*), action recognition (*act*), world knowledge (*know*), spatial reasoning (*spat*), object identification (*ident*), temporal reasoning (*temp*), text/OCR (*text*), comparison (*comp*), and causal reasoning (*cause*); and (2) *entity type*—the primary visual referent: person, animal, object, food, vehicle, place, and other. These two dimensions cross-classify the 1,000 questions and enable the fine-grained alignment analysis in Section 6.3, where we show that human–model prior alignment varies dramatically across operation types (Jaccard 0.93 for yes/no vs. 0.13 for count), a distinction invisible at the aggregate level and unreachable without the taxonomy.

3.4 Human Data Collection

We recruited N=36 participants (15 men, 21 women; age 18–34) through online university community forums. All

procedures were conducted under an IRB-approved protocol (compensation: 10,000 KRW/hour, consistent with standard rates). Participants answered 113 VQA-style questions in a text-only blind setting, relying only on the question text and general world knowledge. Responses could be given in Korean or English and were translated and normalized prior to analysis. After each answer, participants rated their confidence on a 5-point Likert scale. Unless otherwise noted, the human-comparison analyses in Section 6 are matched to the model condition used for those exports rather than assumed to be interchangeable across all blind prompts. Full protocol details are provided in Appendix A.

3.5 Metrics

We report: (1) **VQA accuracy** (standard 10-annotator soft evaluation, Antol et al. 2015), used as context rather than primary evidence; (2) **answer distribution overlap** via Jaccard similarity of top-3 answers per question, measuring shared prior structure independently of correctness; (3) **semantic similarity** via pairwise SBERT cosine, SimCSE cosine, and BERTScore F1 for open-ended answer agreement; (4) **Pearson difficulty correlation:** r between per-question human accuracy $h(q)$ (proportion of 36 participants correct) and per-model accuracy, measuring alignment at the question-difficulty level; (5) **Krippendorff’s α** (nominal scale) for within-group and cross-group inter-rater reliability; and (6) **logprobability confidence**, extracted as the mean token-level log-probability of the generated answer (top-5 log-probs, temperature 0). Note: $h(q)$ uses VQA soft-scores (0/0.33/0.67/1.0 averaged across raters); model accuracy is binary (0 or 1).

For the answer-agreement analyses in Section 6, we compare human-human, human-model, and model-model answer pairs using both **lexical** and **semantic** similarity metrics after shared answer normalization. The lexical metrics are **exact match**, **token Jaccard**, **ROUGE-1**, and **chrF**. These metrics capture progressively softer forms of surface overlap, from identity of the normalized answer string to word- and character-level overlap. The semantic metrics are cosine similarity between **Sentence-BERT** embeddings (Wang et al. 2020), cosine similarity between **SimCSE** embeddings, and **BERTScore F1**. These metrics are intended to capture paraphrastic or near-synonymous answers that lexical overlap misses.

Our main interpretive focus is on the semantic metrics, especially Sentence-BERT cosine, because the central question is not whether models copy the exact human string, but whether different model groups recover the *same human prior* at the meaning level. Lexical metrics remain useful as strict diagnostics, but they are brittle for short VQA answers and can understate agreement when humans and models express the same prior using different wording. We therefore use lexical metrics as supporting evidence and semantic agreement as the primary criterion when asking which model families are closest to human blind reasoning.

4 Linguistic Prior Exploitation: Distribution Biases and Prior Structure

We characterize the *structure* of blind model behavior—what answer distributions models produce, and how those distributions depend on lexical cues in the question. Accuracy serves as secondary context; the primary evidence is the shape of answer distributions and how they compare to human baselines.

Blind VQA accuracy ranges from 35–49% across models (see Appendix D for the full per-model breakdown). Text-only Qwen3 LLMs (no vision encoder) achieve 35–38%, approaching their VLM counterparts, confirming that blind performance is driven by the language backbone. Scale does not monotonically improve blind VQA accuracy, consistent with a ceiling on high-frequency question–answer patterns. The instruction has little effect on accuracy even when it substantially changes answer content (see Appendix E), confirming that prior-driven answering is not corrected by awareness of the missing image.

4.1 Answer Distribution Biases

A key question is not just *whether* models answer correctly under blind conditions, but *how* their answer distributions differ from humans and from ground truth. As shown in Figure 4, three systematic biases emerge:

No-bias in yes/no questions. Under blind conditions, 70% of model yes/no answers are “no” (vs. ground truth: 47% no). Humans show a smaller bias (53% no), suggesting that the model’s extreme no-bias reflects an epistemic default—the null answer under uncertainty—rather than a corpus prior. Notably, this reverses under blind+inst (“yes” increases from 30% to 55%), confirming that the “no” default is instruction-suppressed rather than a genuine linguistic prior.

Zero-bias in count questions. 75% of model numeric answers under blind conditions are “0,” while humans produce a much more distributed count distribution. The interpretation is consistent: in the absence of visual evidence, models default to “there are none.”

Black-bias in color questions. Among open-ended color answers, “black” accounts for 16% of model responses under blind conditions, far exceeding its ground-truth frequency. This likely reflects the modal color in the training distribution for generic object descriptions.

These null-hypothesis biases illustrate that models are not randomly sampling from their prior—they produce a specific failure mode that partially mimics the structure of a reasonable uncertainty response.

5 Characterizing Prior Reliance

The analyses in this section move from *how much* blind performance models obtain to *how* they produce it: whether their priors survive lexical degradation, whether a small instruction changes their willingness to commit, and whether those commitments are calibrated. Figure 1 provides an overview: the left panel shows the per-question structure of human–VLM alignment (human accuracy vs. model accuracy, colour-coded by answer overlap), and the right panel

shows how model accuracy degrades across control variants while human accuracy remains nearly flat.

5.1 Control Question Degradation

Figure 1 (right) reveals the core finding: as entity-specific lexical content is progressively removed—from the original question (C) to a hypernym-weakened variant (B) to a fully pronominalized form (A)—model accuracy drops substantially for all groups (VLM: 0.455 $\rightarrow$ 0.364, backbone decoder: 0.506 $\rightarrow$ 0.332), while human accuracy decreases by only 0.035 (0.262 $\rightarrow$ 0.227). The asymmetric degradation confirms that model accuracy under blind conditions is driven by entity-specific corpus-frequency priors: removing the entity name (“dog”) for a pronoun (“it”) collapses the prior signal, whereas humans can reason about the abstract question structure regardless. The backbone decoder shows the steepest absolute drop ($\Delta = 0.174$), suggesting its blind advantage is particularly reliant on entity-specific n-gram co-occurrence rather than general syntactic patterns. Fine-grained per-entity-type degradation curves are in the Appendix.

5.2 Instruction-Gated Hallucination

A key question is whether models’ blind answers reflect a readily available prior that is always active, or a latent prior that requires explicit permission to express. We investigate this by comparing model outputs between the **Blind** and **Blind+Inst** conditions.

We classify model outputs in the blind condition as: *hard-abstained* (explicit refusal: “I cannot answer”, “no image”); *soft-abstained* (hedging outputs: “nothing”, “none”, “unanswerable”, “nowhere”); *hallucinated_correct* (specific answer matching GT); or *hallucinated_wrong* (specific answer not matching GT). Hard abstentions are near-zero across all models (0.0–0.3%), consistent with the baked-in “Answer using a single word” instruction suppressing explicit refusals. The real signal is *soft abstention*: 11–19% of blind outputs are hedging responses, varying by model family.

Figure 2 shows the *soft abstention collapse rate*: the fraction of soft-abstained blind responses that convert to specific answers under the Blind+Inst instruction. This rate varies dramatically:

Model	Soft-abs (blind)	Collapse rate
Qwen3-VL-8B	12.9%	71.9%
LLaVA v1.6-Mistral	15.8%	51.9%
LLaVA v1.6-Vicuna	11.3%	28.3%
LLaVA v1.5	18.7%	17.1%

We interpret this spectrum as measuring the degree to which prior activation is *instruction-gated*. Qwen3-VL, a highly instruction-tuned model, largely suppresses its prior under blind conditions but rapidly activates it when given permission. LLaVA v1.5, by contrast, hallucinates *unconditionally*: it produces specific answers whether or not instructed to, and shows almost no response to the permission instruction. This distinction is mechanistically important: instruction-gated models have a functioning “uncertainty” signal that instruction can override, while unconditional models lack this gate entirely.

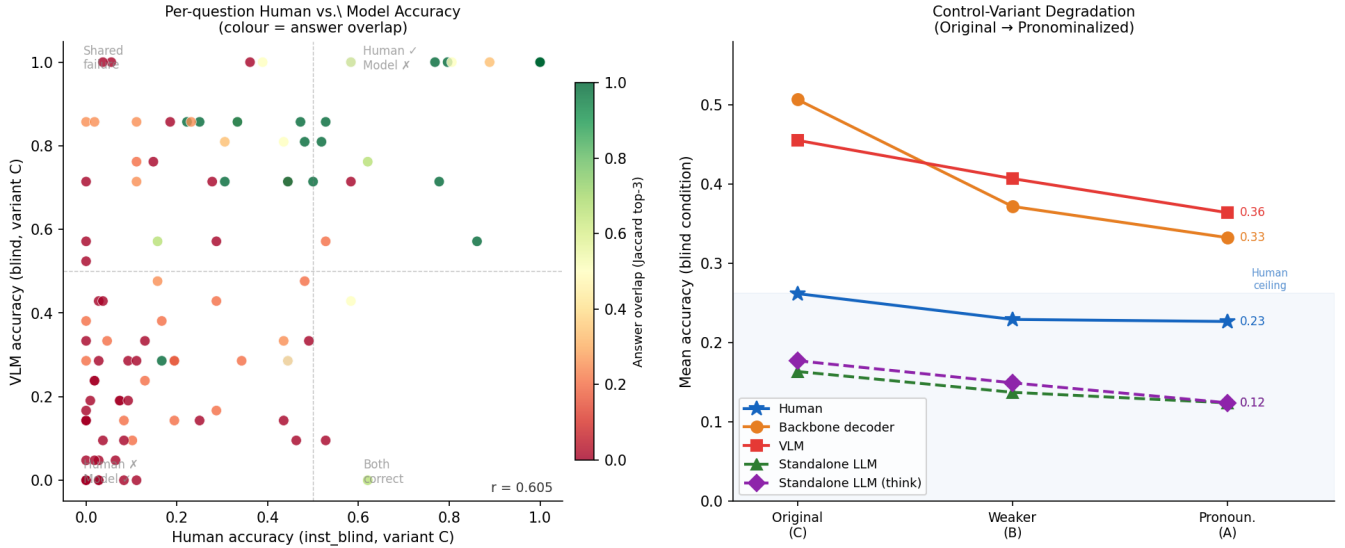


Figure 1: **Overview of prior exploitation and human-model alignment.** **Left:** per-question human accuracy (x-axis, inst. blind) vs. VLM accuracy (y-axis, blind), colour = Jaccard answer overlap (top-3). Green dots indicate shared correct answers (both parties exploit the same prior); red dots indicate divergence. The overall correlation is $r = 0.605$. **Right:** mean accuracy across control variants (Original → Weaker → Pronominalized), blind condition. All model groups degrade substantially as entity anchors are removed, while human accuracy (blue) remains nearly flat ($0.26 \rightarrow 0.23$). This contrast confirms that model performance is driven by entity-specific lexical priors rather than structural question understanding.

Additionally, 33.6% of all answers (not just soft-abstentions) change between blind and blind+inst conditions for Qwen3-VL-8B—a large behavioral shift driven by a single sentence. The yes/no distribution shifts from 70% “no” in blind to 55% “yes” in blind+inst, consistent with the instruction activating a “plausible scenario exists” prior that overrides the default epistemic null.

5.3 Confidence Calibration

Logprobability analysis reveals that blind hallucinations are produced with extreme confidence. For Qwen3-VL-8B under blind conditions, 80.8% of output tokens have mean logprob > -0.5 ($\approx 60\%$ probability per token), and the mean logprob across all blind outputs is -0.258 . Under Blind+Inst, mean logprob rises to -0.177 and the fraction with logprob > -0.5 increases to 89.5%. The instruction does not merely change *which* answer is produced—it also makes the model more *certain* about whatever it produces.

The few soft-abstained outputs (“nothing”, “none”) have lower mean logprob (-0.37 vs. -0.26 for answered outputs), confirming that hedging outputs reflect genuine uncertainty. However, the effect size is small, and the baseline level of confidence even for soft-abstentions is high. The primary calibration failure is not that models are uncertain when they should be confident, but that they are *highly confident about wrong hallucinated answers* they should be uncertain about.

Quantifying this overconfidence with Expected Calibration Error and testing the corpus-frequency correlation (McCoy, Frank, and Linzen 2024) directly are reserved for future work.

6 Human-Model Alignment

Figure 3 summarises our central finding: when models answer VQA questions without images, their outputs are surprisingly close to human blind responses. VLM backbone decoders (the language backbone extracted without the vision encoder) reach the highest human-model SBERT similarity (0.501), within 0.067 of the human-human ceiling (0.568). Full VLMs cluster nearby (0.480), while standalone LLMs span a wide range—Mistral-7B and Vicuna-13B diverge sharply (SBERT < 0.25 , exact match ≈ 0), whereas instruction-tuned models such as Qwen2.5-7B and Qwen3 series approach the VLM cluster.

The key point of this section is architectural rather than purely descriptive: the model component that best recovers human blind priors is the language backbone exposed in the decoder-only setting, not the full vision-language stack. The question-type analysis that follows explains where this alignment is strongest, but the primary result is the model-group separation itself.

6.1 Humans as a Reference for Prior Structure

Human participants answering without images still rely on world knowledge and linguistic priors, but they are not shaped by the same corpus-frequency objective that drives pretrained LLMs. This makes human blind behavior a reference for what *linguistically natural* uncertainty looks like—as opposed to what the training distribution would predict.

The contrast is sharpest in answer distributions (Figure 4). For yes/no questions, VLMs default to “no” 62% of the time, while humans lean toward “yes” 67%—the opposite direction. For count questions, 70% of VLM answers are

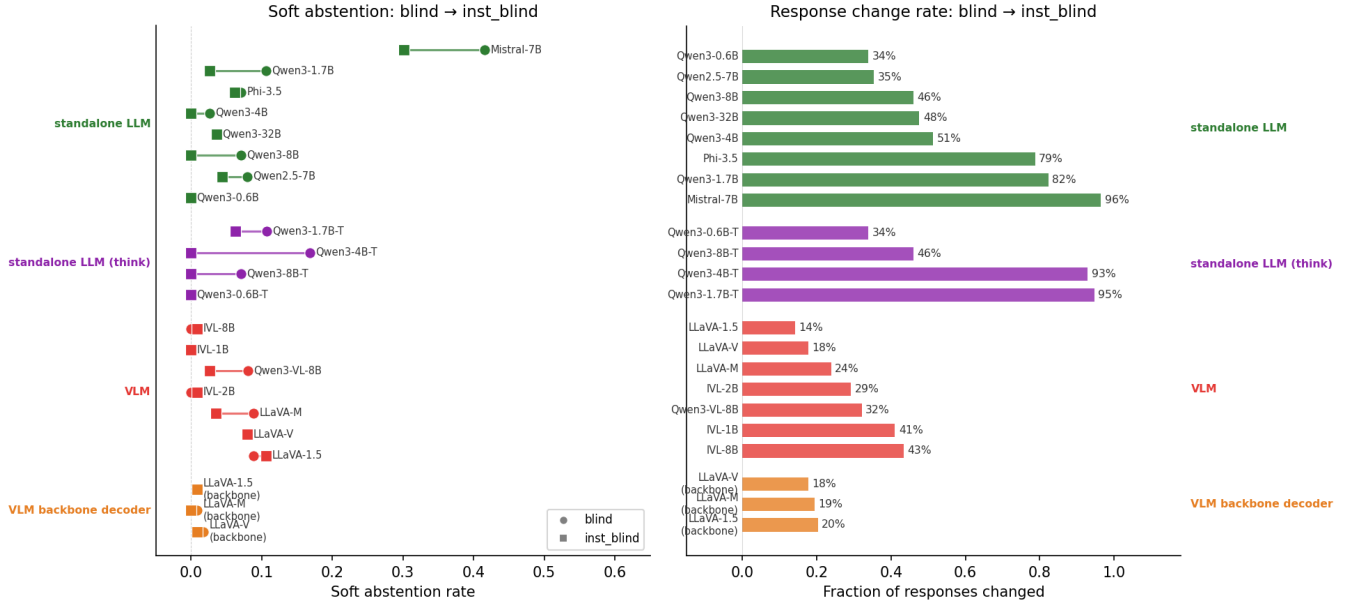


Figure 2: Effect of the Blind+Inst instruction across all models (variant C). **Left:** soft abstention rate (hedging outputs such as “nothing”, “none”, “nowhere”) before (circle) and after (square) instruction. Most models reduce hedging, but the collapse is sharpest for thinking-mode LLMs and instruction-tuned VLMs; backbone decoders remain near zero throughout. Mistral-7B maintains high soft abstention (41%→30%) regardless. **Right:** fraction of responses that change between blind and inst_blind conditions. Standalone LLMs are most instruction-sensitive (34–96% responses change); backbone decoders are the most stable (18–20%), indicating that their outputs are driven primarily by the question’s linguistic prior rather than by the instruction framing.

“0” while humans produce a distributed count distribution (modal answer: 2, followed by 1 and 3). This reversal is not a calibration gap in the same direction; it indicates that the model prior is qualitatively different from the human prior for these question types.

6.2 Which Models Are Most Human-Like?

Not all model families align equally with human blind behavior. We compare four groups—VLMs, VLM backbone decoders (the same language backbone without the vision encoder), standalone LLMs, and thinking-mode LLMs—using both difficulty correlation and pairwise answer similarity. The main interpretive question is whether the model component that best matches humans is the visual stack, the language backbone, or the instruction-tuning regime layered on top.

Pairwise answer similarity. We compute pairwise SBERT cosine similarity (Wang et al. 2020) between all answer pairs (human-human, human-model, model-model) across the 374 free-text control questions. Human-human agreement provides the ceiling: $\overline{\text{SBERT}}_{\text{HH}} = 0.568$. VLM backbone decoders reach 0.501—closest to the human ceiling—while full VLMs score 0.480, standalone LLMs 0.398, and thinking-mode LLMs 0.426. The gap between backbone decoders and full VLMs ($\Delta = 0.021$) is consistent across lexical metrics (exact match, chrF) and semantic metrics (SimCSE and BERTScore), indicating that

removing the vision encoder does *not* hurt human alignment and may even improve it by reducing the model’s reliance on spurious visual features.

Difficulty correlation by model group. Pearson correlation between per-question human accuracy $h(q)$ and per-model accuracy further separates the groups. Under blind conditions, VLMs correlate at $r = 0.50$ – 0.62 ; backbone decoders at $r = 0.58$ – 0.68 ; standalone LLMs at $r = 0.35$ – 0.55 . The Blind+Inst condition consistently raises correlations by up to $\Delta r = +0.10$ for VLMs, narrowing the gap to the human reference. This pattern is consistent with the instruction-gated hallucination result (Section 5.2): receiving the “imagine the scene” instruction pushes models to activate the same distributional priors that drive human performance on these questions.

Inter-rater structure. The inter-rater agreement matrix (Figure 7) shows that human participants cluster separately from all model groups, confirming a qualitative difference in reasoning strategy rather than a mere accuracy gap. Backbone decoders sit closest to the human cluster, standalone LLMs furthest. Krippendorff’s α decreases when any model is added to the human rater pool (from 0.554 to 0.535–0.543), but the decrease is smallest for backbone decoders, reinforcing that the language backbone is the primary driver of human-like prior structure in VLMs.

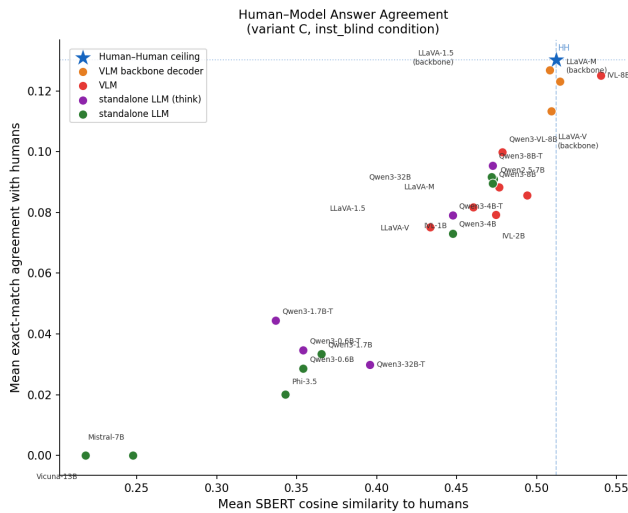


Figure 3: Per-model human-model answer agreement (inst.blind, variant C). X-axis: mean SBERT cosine similarity to human answers. Y-axis: mean exact-match agreement with humans. The blue star marks the human-human ceiling. Backbone decoders (orange) cluster closest to humans; standalone LLMs (green) vary widely, with chat-tuned models (Qwen3, Qwen2.5) near the VLM cluster and base-style models (Mistral, Vicuna) far below. The outlier InternVL-8B (red) exceeds the human SBERT ceiling, suggesting its language backbone has particularly strong alignment with human linguistic priors.

6.3 Shared vs. Divergent Priors by Entity Type

Not all questions trigger the same degree of human-model prior alignment. The entity taxonomy introduced in Section 3.3 provides the analytical structure: as shown by the t-SNE embedding visualization (Appendix F), entity groups form coherent semantic clusters in the question embedding space, making entity type a principled grouping dimension for alignment analysis.

We measure **difficulty correlation**—Pearson r between per-question human accuracy $h(q)$ and VLM accuracy—within each of the 9 entity types across 113 questions. Figure 5 shows the distribution of questions by entity type (left) and the resulting correlations (right).

Three alignment regimes emerge:

Strong alignment (animal, place). Animal and place questions show near-perfect human-model difficulty correlation ($r = 0.90, 0.91$). Both humans and models exploit the same entity-specific corpus frequencies for these categories—questions about animals or locations tend to have predictable answers that align with statistical regularities in training data.

Moderate alignment (food, person, other). Food ($r = 0.77$) and person ($r = 0.64$) entities show moderate alignment. Models capture the broad easy/hard structure of human difficulty but diverge on ambiguous cases, particularly for person questions where identity and contextual knowl-

edge interact.

Divergence (object, product). Object questions show weak alignment ($r = 0.13$), reflecting the high diversity of object categories that span many prior strengths. Product questions are anti-aligned ($r = -0.48$): when humans succeed on product queries, models tend to fail, likely because product-specific prior knowledge is sparser in model training data than in human experience.

Figure 6 shows the per-question scatter. The majority of questions lie above the diagonal (models answer more questions correctly than humans under blind conditions), and the green dots—high Jaccard overlap—cluster near the origin (both wrong) and near (1,1) (both correct), confirming that prior sharing is strongest when both parties are equally uncertain or equally knowledgeable. The anti-aligned count questions (dark red dots) are visible as a cloud where human accuracy is spread but model accuracy is near zero (zero-bias) or near one.

6.4 Human-Model Divergence in Answer Patterns

We measure human-model alignment using two complementary metrics. First, we compute **Pearson difficulty correlation**: for each question q , we compute the human group accuracy $h(q)$ (proportion of 36 participants correct) and correlate it with per-model accuracy. We prefer Pearson over Spearman here because the relationship is approximately linear (max $|r - \rho| = 0.026$ across all conditions), and operating on proportions avoids the information loss from rank conversion on binary data. Second, we compute **Krippendorff’s α** (nominal scale, binary correctness) to measure within-group and cross-group reliability.

Human difficulty estimates are highly reliable: $\text{ICC}(2, 36) = 0.963$ across 36 raters, justifying use of $h(q)$ as a stable difficulty signal. Within-group human reliability is $\alpha_{\text{human}} = 0.554$ (moderate, appropriate for a genuinely ambiguous task without visual input).

Pretrained VLMs show moderate correlation with human difficulty under blind conditions (Pearson $r = 0.50$ – 0.62 depending on model), confirming that human and model difficulty patterns partially overlap—both exploit linguistic priors to some extent. However, adding any model to the human rater pool *decreases* α (from 0.554 to 0.535–0.543), confirming that models are structurally less consistent with the human group than humans are with each other.

In the current analysis, Blind+Inst often yields slightly higher human-difficulty correlations than Blind alone for several model families. This pattern is consistent with the soft-abstention-collapse result in Section 5.2, but it should be treated as provisional until the human comparison conditions are fully matched across settings.

The group-mean SBERT heatmap (Figure 7) quantifies the structural separation: human-human similarity (0.568) exceeds all cross-group scores, confirming that humans form a distinct cluster. VLM backbone decoders reach the highest human alignment (0.501), followed by thinking-mode LLMs (0.426) and full VLMs (0.480). Standalone LLMs score lowest (0.398). The pattern is consistent across

Figure 4: Answer distribution biases under blind conditions, across all model groups (Ground Truth, Humans, VLM, VLM Decoder, Standalone LLM, Standalone LLM (think)). **Left (yes/no):** VLMs default to “no” ($\approx 62\%$) while humans lean “yes” ($\approx 67\%$)—opposite directions. Backbone decoders are more balanced; standalone LLMs are the most no-biased. **Right (count/number):** VLMs collapse to “0” ($\approx 70\text{--}75\%$) while humans produce a distributed count distribution (modal: 2, followed by 1 and 3). Both biases are absent in ground truth and diverge from all human patterns, confirming they are model-specific artifacts rather than shared linguistic priors. (Generated by notebook 13.)

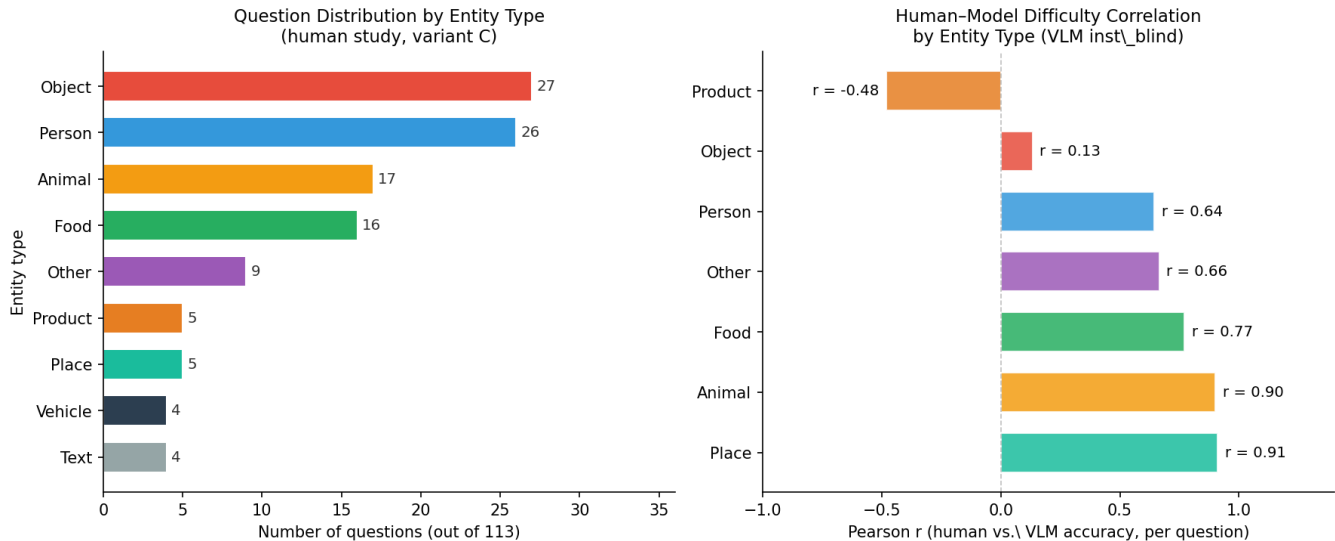


Figure 5: **Left:** distribution of 113 study questions by entity type. Object and person dominate; place, product, text, and vehicle have small samples ($n \leq 5$). **Right:** Pearson r between per-question human accuracy and VLM accuracy (inst.blind, variant C) within each entity type. Animal ($r = 0.90$) and place ($r = 0.91$) questions show near-perfect human–model alignment; food and person show moderate alignment ($r = 0.77, 0.64$). Object questions show weak alignment ($r = 0.13$), while product questions are anti-aligned ($r = -0.48$), indicating that human and model difficulty patterns diverge for product-related queries.

all seven metrics (exact match, Jaccard, ROUGE-1, chrF, SBERT, SimCSE, BERTScore); full per-rater heatmaps for each metric are provided in the Appendix.

Semantic similarity analysis reinforces this finding: model answers have higher embedding similarity to ground truth (mean 65.8 vs. human 64.2) despite being generated without visual evidence, and the human similarity distribution has a longer left tail (more diverse wrong answers), consistent with exploratory reasoning rather than prior retrieval.

7 Discussion

The instruction-gated spectrum as an architectural insight. The dramatic variation in soft-abstention collapse rates (17% for LLaVA v1.5 vs. 72% for Qwen3-VL) suggests that instruction-following capability plays a role in prior activation that goes beyond accuracy. Instruction-tuned models have learned to gate prior expression behind contextual permission signals, while models with weaker instruction following leak priors unconditionally. This is not simply a capability difference—LLaVA v1.5 is competitive on standard VQA accuracy—but a behavioral difference in how priors are accessed.

Training-free intervention directions. Our findings motivate two training-free interventions. First, *confidence-based selective abstention*: since overconfident blind answers cluster on known bias patterns (“no” for yes/no, “0” for count), setting a per-question-type logprob threshold to withhold low-information answers can reduce prior-driven outputs without any model update. Second, *conditional abstention prompting*: replacing the instruction “imagine the scene” with “only answer if you can do so without visual information” inverts the permission signal and tests whether models have self-knowledge of prior reliance—high soft-abstention collapse rate under the standard instruction (Qwen3-VL-8B: 67%) suggests models have a functioning uncertainty gate that can be redirected. Unlike finetuning-based approaches, both interventions operate at inference time, are architecture-agnostic, and directly use the behavioral signals already characterized in this study.

Implications for benchmark design. Our results suggest that any VQA benchmark can be partially solved by linguistic priors alone, and that the magnitude of this effect varies predictably with question type, answer distribution, and entity specificity. We propose that blind VQA evaluation should become a standard diagnostic for new VQA benchmarks: a high blind accuracy ($>40\%$) is at least a warn-

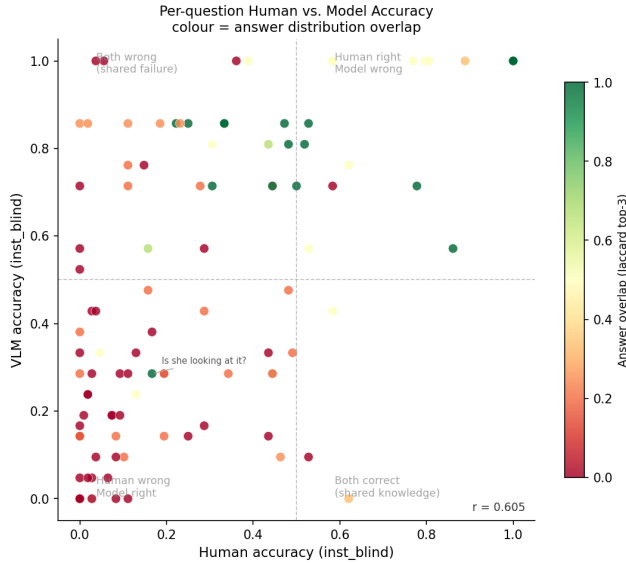


Figure 6: Per-question human accuracy (x-axis) vs. VLM accuracy (y-axis, inst_blind, variant C). Colour encodes answer distribution overlap (Jaccard top-3). Green points (top-left and bottom-left quadrants) are questions where both parties fail similarly—shared language priors operating without visual grounding. The overall correlation is $r = 0.605$.

ing sign that the benchmark may over-index on knowledge-accessible question types.

Limitations. Our human study involves 36 participants recruited from university communities in a single language (Korean/English), limiting cultural and demographic generalizability. Model evaluation uses greedy decoding (temperature 0), which may underestimate answer diversity and overestimate confidence. The control variant GPT extraction uses a best-effort seed (seed is advisory in the Responses API) and results may vary slightly across runs. Human data was collected under the Blind+Inst condition only, so direct comparisons with model blind (no instruction) behavior rely on the assumption that the instruction has a comparable normalizing effect on humans and models—an assumption our instruction-gated analysis partially supports but does not fully validate. A direct corpus-frequency correlation—testing whether per-question blind accuracy predicts answer frequency in the VQA v2 training distribution—would strengthen the mechanistic account (McCoy, Frank, and Linzen 2024) and is deferred to future work.

8 Conclusion

We present a human-calibrated diagnostic study of linguistic prior exploitation in Vision-Language Models. Analogous to hypothesis-only baselines in NLI, blind VQA reveals that VLMs produce characteristic answer distribution biases shaped by training-corpus frequencies—and that these biases are only partially shared with human blind reasoners. We introduce control question variants to probe how lexical cues drive prior activation, and use logprob analysis to ex-

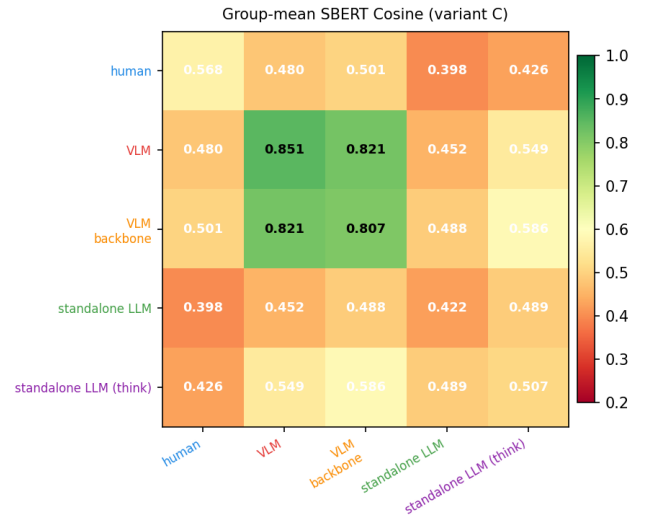


Figure 7: Group-mean SBERT cosine similarity between all rater pairs (variant C). Diagonal = within-group similarity; off-diagonal = cross-group. VLM backbone decoders (language backbone only, no vision encoder) are the closest to the human block (0.501), while standalone LLMs are furthest (0.398). Full per-rater heatmaps for all metrics are in Appendix.

pose the severe overconfidence that accompanies blind hallucination. The instruction-gated spectrum reveals a behavioral axis across model families that is invisible to accuracy-only evaluation.

The central finding of the human alignment study is that prior sharing is question-type-specific. Yes/no and action questions show near-complete human-model overlap (Jaccard ≈ 0.93), while count and grounded-reference questions are anti-aligned (difficulty $r = -0.39$). The model zero-bias and no-bias are not linguistic priors in any human sense—they are model-specific failure modes. At the architecture level, VLM backbone decoders (language backbone without vision encoder) achieve near-human pairwise answer similarity (0.501 vs. human 0.568), and their yes/no distribution reverts to human-like rates when the vision encoder is removed, confirming that the no-bias is a VLM artifact rather than a corpus-frequency prior. These findings argue that diagnosing prior exploitation requires attending to both the language backbone and the vision-language integration layer separately. Correcting it need not require retraining; confidence-based selective abstention and conditional abstention prompting offer inference-time handles that are directly motivated by the behavioral signals characterized here.

References

- Agrawal, A.; Batra, D.; Parikh, D.; and Kembhavi, A. 2018. Don’t just assume; look and answer: Overcoming priors for visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Antol, S.; Agrawal, A.; Lu, J.; Mitchell, M.; Batra, D.; Zit-

nick, C. L.; and Parikh, D. 2015. VQA: Visual question answering. In *Proceedings of the IEEE International Conference on Computer Vision*, 2425–2433.

Bai, S.; Cai, Y.; Chen, R.; Chen, Z.; Deng, L.; Ding, W.; Gao, C.; Ge, C.; Ge, W.; Guo, Z.; et al. 2025. Qwen3-VL Technical Report. *arXiv preprint*.

Chen, L.; Li, J.; Dong, X.; Zhang, P.; Zang, Y.; Chen, Z.; Duan, H.; Wang, J.; Qiao, Y.; Lin, D.; et al. 2024. Are we on the right way for evaluating large vision-language models? *Advances in Neural Information Processing Systems*, 37: 27056–27087.

Goyal, Y.; Khot, T.; Summers-Stay, D.; Batra, D.; and Parikh, D. 2016. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. *International Journal of Computer Vision*.

Gururangan, S.; Swayamdipta, S.; Levy, O.; Schwartz, R.; Bowman, S.; and Smith, N. A. 2018. Annotation artifacts in natural language inference data. In *Proceedings of NAACL-HLT 2018*, 107–112.

Han, T.; Lian, Q.; Pan, R.; Pi, R.; Zhang, J.; Diao, S.; Lin, Y.; and Zhang, T. 2024. The instinctive bias: Spurious images lead to illusion in MLLMs. *arXiv preprint arXiv:2402.03757*.

Ji, Z.; Lee, N.; Frieske, R.; Yu, T.; Su, D.; Xu, Y.; Ishii, E.; Bang, Y. J.; Madotto, A.; and Fung, P. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12): 1–38.

Lee, K. i.; Kim, M.; Yoon, S.; Kim, M.; Lee, D.; Koh, H.; and Jung, K. 2024. VLind-Bench: Measuring language priors in large vision-language models. *arXiv preprint*.

Liu, H.; Li, C.; Li, Y.; Li, B.; Zhang, Y.; Shen, S.; and Lee, Y. J. 2024. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. *arXiv preprint*.

Liu, H.; Li, C.; Wu, Q.; and Lee, Y. J. 2023. Visual instruction tuning. *Advances in Neural Information Processing Systems*, 36: 34892–34916.

McCoy, R. T.; Frank, R.; and Linzen, T. 2024. Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences*, 121(41): e2322420121.

OpenAI. 2025. GPT-4o System Card. Technical report, OpenAI. <https://openai.com/index/gpt-4o-system-card>.

Poliak, A.; Naradowsky, J.; Haldar, A.; Rudinger, R.; and Van Durme, B. 2018. Hypothesis only baselines in natural language inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, 180–191.

Rohrbach, A.; Hendricks, L. A.; Burns, K.; Darrell, T.; and Saenko, K. 2018. Object hallucination in image captioning. In *Conference on Empirical Methods in Natural Language Processing*.

Tversky, A.; and Kahneman, D. 1974. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157): 1124–1131.

Wang, W.; Wei, F.; Dong, L.; Bao, H.; Yang, N.; and Zhou, M. 2020. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in Neural Information Processing Systems*, 33: 5776–5788.

Xiao, Y.; and Wang, W. Y. 2021. On hallucination and predictive uncertainty in conditional language generation. In *Conference of the European Chapter of the Association for Computational Linguistics*.

Yang, A.; Li, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Gao, C.; Huang, C.; Lv, C.; et al. 2025. Qwen3 Technical Report. *arXiv preprint*.

A Human Study Protocol

A total of 36 participants (15 men and 21 women; age range: 18–34 years) were included in the study. Participants were recruited via online student community forums at universities. All procedures were conducted under an IRB-approved protocol and adhered to the principles of the Declaration of Helsinki. Informed consent was obtained from all participants prior to participation.

Each participant completed 113 VQA-style questions. Participants were compensated at a rate of 10,000 KRW per hour, consistent with standard compensation for similar annotation tasks.

For VQA, participants were instructed to provide free-form answers that were as concise as possible. Responses could be given in either Korean or English and were translated and processed using the same answer normalization and aggregation strategy described in Appendix B.

B Answer Aggregation

We cluster semantically equivalent answers (e.g., “3” and “three”) into canonical forms using GPT-4o mini (OpenAI 2025). The prompt instructs the model to group synonymous responses, select the most concise canonical form, and return results in structured JSON. Human responses are then aggregated into a confidence-weighted distribution over canonical answers, where each answer’s weight is the sum of confidence scores from all annotators selecting that answer, normalized to sum to 1:

$$H(a_k^*) = \frac{\sum_{i: \phi(a_i) = a_k^*} c_i}{\sum_i c_i} \quad (1)$$

where c_i is participant i ’s Likert confidence (normalized to $[0, 1]$), ϕ maps raw answers to canonical answers, and K is the number of canonical answer clusters.

C Control Question Extraction Prompt

Control variants are extracted using GPT-4.1-mini-2025-04-14 (temperature 0, seed 42) with strict JSON schema mode for reproducibility. The system prompt instructs the model to generate all four variants in a single API call per batch of 50 questions. The schema enforces: `{deictic_removed, object_removed, weaker_object, pronominalized}` as required string fields, with `additionalProperties: false`. For pronominalized, the prompt specifies grammar-aware

substitution rules to ensure natural pronominalization (“What color is it?” not “What color is the object?”).

D Blind VQA Accuracy Summary

E Blind Awareness Instruction

After the original prompt and the blank image, models in the Blind+Inst condition receive: “*Note: No images are provided. For each question, imagine an appropriate image exists and answer based on the most common or universal scenario.*”

Table 2 shows accuracy comparison between Blind and Blind+Inst conditions on VQA. The instruction usually changes accuracy by only a few points, confirming that it does not reliably improve performance even when it substantially changes answer behavior.

F Question Embedding Space (t-SNE)

Figure 8 shows the t-SNE projection of all 1,000 VQA questions using sentence embeddings (all-MiniLM-L6-v2), with the 113 sampled study questions highlighted as large dots. The three panels colour points by entity group, operator group, and question word respectively.

Entity groups (left panel) form coherent spatial clusters—animals, persons, and places each occupy distinct regions of the embedding space. Operator groups (center panel) are more interleaved, with attribute and count questions distributed throughout. This structural difference motivates our choice to analyse human–model alignment by entity type rather than operation type: entity boundaries are semantically well-defined and correspond to distinct prior structures, whereas operation types are cross-cutting and do not form natural semantic partitions.

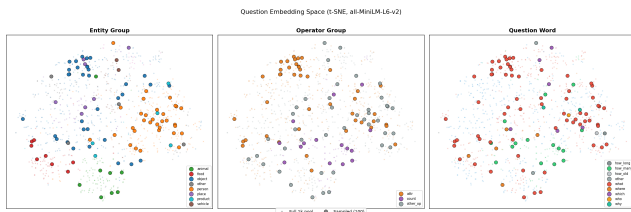


Figure 8: t-SNE of 1,000 VQA question embeddings. Large dots = 113 sampled study questions. **Left:** entity group—clear spatial clustering by entity type motivates entity-based alignment analysis. **Centre:** operator group—attr, count, and other_op are interleaved, not separable. **Right:** question word—what, where, who, how_many overlap substantially. Entity type is the most structurally informative grouping.

G Answer Distribution by Model Group (Notebook 13)

Figures 9 and 10 show normalized answer distributions for yes/no and count questions respectively, broken down by source (Ground Truth, Humans, VLM, VLM decoder, Standalone LLM, Standalone LLM (think)) under both the Blind and Blind+Inst conditions. These correspond to Figure 4 in

the main text; the inst_blind versions are shown here for comparison.

Figure 9: Yes/No answer distribution by source group. Left: Blind condition. Right: Blind+Inst condition. VLMs default to “no” ($\approx 62\%$) while humans lean “yes” ($\approx 67\%$), the opposite direction. Backbone decoders are more balanced. Standalone LLMs are heavily biased toward “no” under blind conditions but shift substantially under instruction. (Generated by notebook 13.)

Figure 10: Count/number answer distribution by source group. Left: Blind condition. Right: Blind+Inst condition. VLMs collapse to “0” ($\approx 75\%$) while humans produce a distributed count distribution (modal: 2). Backbone decoders and standalone LLMs also show strong zero-bias. The instruction reduces zero-bias only marginally across all model groups. (Generated by notebook 13.)

Table 1: Zero-shot blind VQA performance of humans and models. Y/N = yes/no accuracy; Num = numerical; Other = open-ended (Accuracy/Similarity). Bold: best; underline: second best.

Model	LLM	Size	Y/N	Num	Other	Avg
Humans (N=36)	–	–	66.6	13.8	13.0/45.5	39.0
<i>LLM Baseline (text-only)</i>						
Qwen3	Qwen3	4B	54.9	19.0	24.7/55.2	38.8
		8B	50.3	17.1	22.1/53.2	35.3
<i>VLM (blind)</i>						
Qwen3-VL	Qwen3	2B	71.8	17.1	23.8/55.5	46.4
		4B	71.5	19.0	28.9/56.4	48.0
		8B	71.5	21.9	27.0/54.9	48.0
LLaVA v1.5	Vicuna	7B	70.3	17.1	19.4/48.4	43.8
LLaVA v1.6	Mistral	7B	69.6	20.0	27.0/55.0	47.0
LLaVA v1.6	Vicuna	7B	74.4	20.0	27.2/56.2	<u>49.4</u>

Table 2: VQA accuracy comparison with and without blind instruction. $\Delta_{\text{inst}} = \text{Blind}_{\text{inst}} - \text{Blind}$.

Model	GT	Blind	Δ
InternVL3.5-1B	80.5	43.6	0.2
InternVL3.5-2B	82.1	45.5	-0.2
InternVL3.5-4B	83.0	45.0	2.2
InternVL3.5-8B	86.9	47.1	-1.7
Qwen3-VL-2B	83.0	45.0	0.7
Qwen3-VL-4B	85.9	45.5	1.9
Qwen3-VL-8B	89.4	44.3	3.1